

System Design Document

Campus Marketplace Application

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1. Introduction

Now in Phase III of the Software Development Life Cycle, design.

1.1 Purpose of Document

The purpose of this document is to describe how the Campus Marketplace Application will be designed and constructed. It defines the system's user interfaces, database structure, and program components.

This document will serve as a reference for developers by providing a clear and detailed project design.

1.2 System Overview

Users will be able to create their own accounts using their official KPU email addresses. The system will allow users to search for product names, tags, categories, etc. Users will also be able to comment on listings once logged in.

1.3 Scope of Design Document

This design document covers the major technical and structural components of the Campus Marketplace Application. It includes:

- The conceptual data model and entity-relationship diagram
- The logical relational schema
- Database normalization and physical database design
- User-interface designs and navigation structure
- Program structure and processing logic

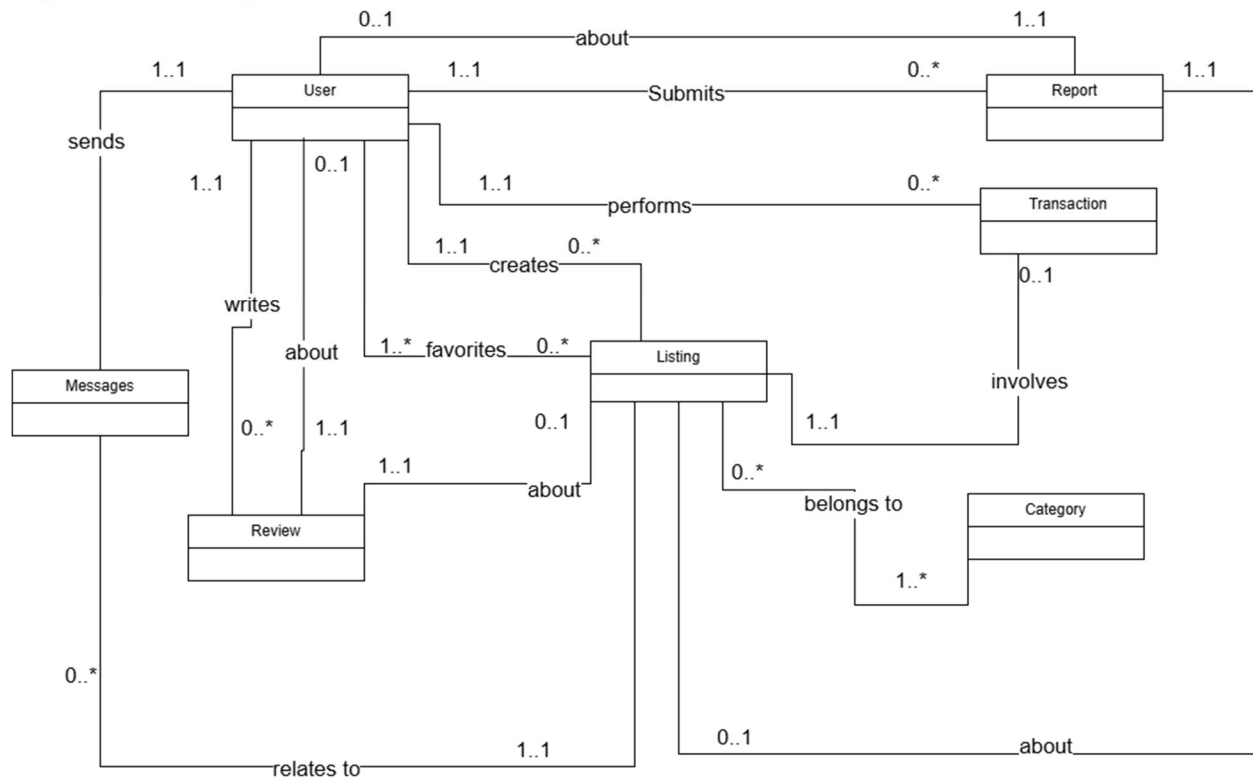
The document focuses on the design of the application and does not provide the complete source code or final deployed system.

2. Conceptual Data Model

This section presents the conceptual data model for the Campus Marketplace Application and provides an explanation of its main components and relationships. The model is used to represent the business processes and information requirements of the Campus Marketplace Application system.

2.1 Conceptual ER Diagram

Figure 2.1: Conceptual ERD



This diagram shows the conceptual relationships among the entities, including basic relationship descriptions and their cardinalities.

2.2 Entity Descriptions

User -

The User entity represents a user account. It stores information about individuals who interact with the system.

Listing -

The Listing entity represents a product uploaded by a seller. It stores information about products.

Category -

The Category entity represents the types of products that can be listed in the Campus Marketplace Application system. It stores all product categories used for classification and search purposes.

Review -

The Review entity represents feedback provided by a buyer on a listing. It stores the content of a single review message submitted by a buyer.

Messages -

The Message entity represents communication history between a buyer and a seller regarding a listing. It stores messages exchanged between two users within the context of a listing.

Report -

The Report entity represents a report submitted by a user. It stores reports related to bugs, illegal activities, or unethical behavior within the system.

Transaction -

The Transaction entity represents a completed transaction between a buyer and a seller. It stores details of transactions.

2.3 Relationship Descriptions

User creates Listing.

Listing belongs to Category.

User writes a Review about Listing.

User sends Messages relates to Listing.

User submits Report about User.

User submits Report about Listing.

User performs Transaction.

Transaction involves Listing.

2.4 Cardinality and Degree Explanation

User – Listing (Binary)

User to Listing is Optional Many relationship.

Listing to User is Mandatory One relationship.

Listing – Category (Binary)

Listing to Category is Mandatory Many relationship.

Category to Listings is Optional Many relationship.

User – Review (Binary)

User to Review is Optional Many relationship.

Review to User is Mandatory One relationship.

User – Messages - Listing (Ternary)

Messages to User is Mandatory One relationship.

Messages to Listing is Mandatory One relationship.

User to Messages is Optional Many relationship.

Listing to Messages is Optional Many relationship.

User – Report (Binary)

User to Report is Optional Many relationship.

Report to User is Mandatory One relationship.

User – Transaction (Binary)

User to Transaction is Optional Many relationship.

Transaction to User is Mandatory One relationship.

Transaction – Listing (Binary)

Transaction to Listing is Mandatory One relationship.

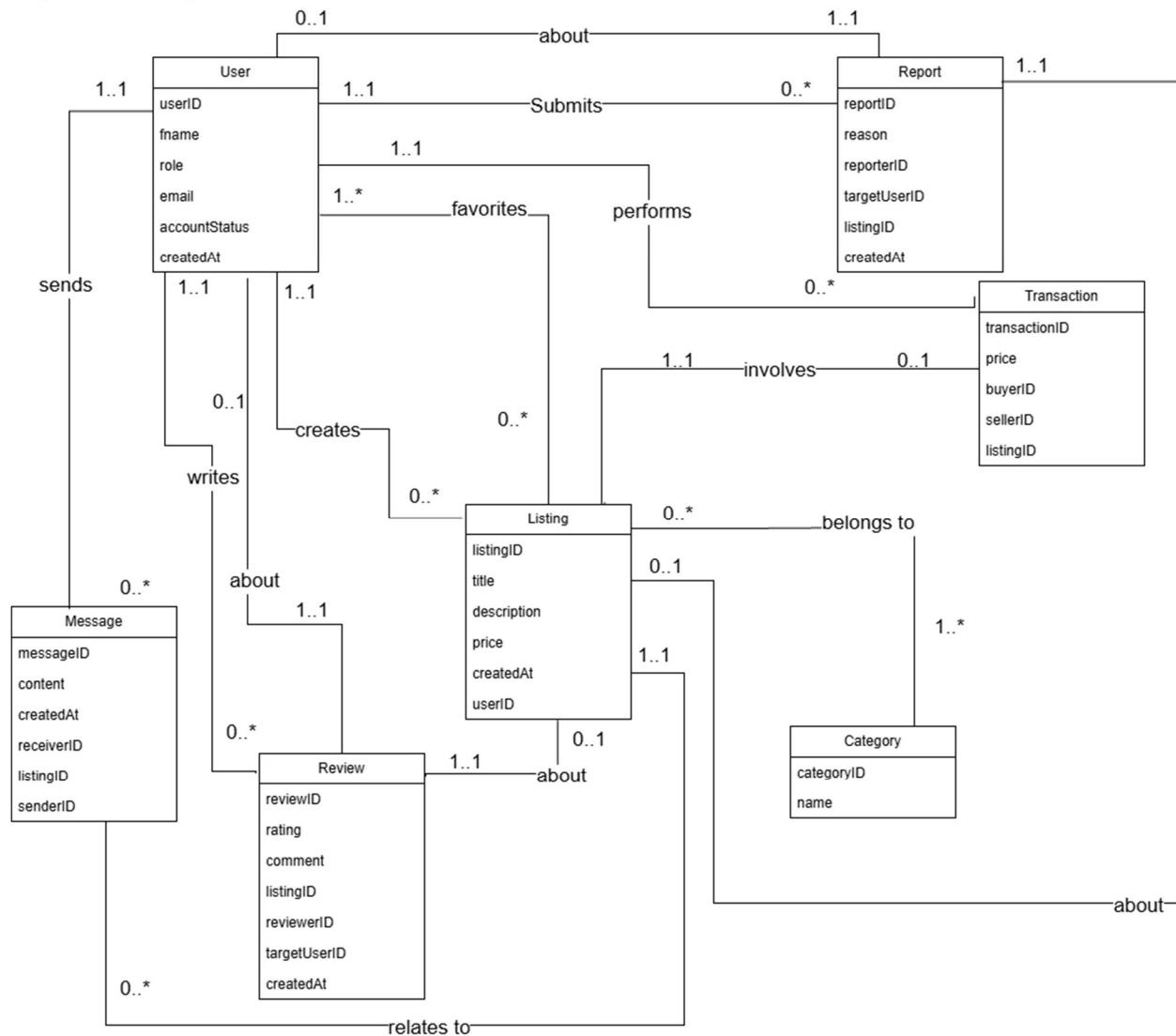
Listing to Transaction is Optional One relationship.

3. Logical Data Model

This section presents the logical data model for the Campus Marketplace Application and explains its relationships and Keys.

3.1 Logical ER Diagram

Figure 3.1: Logical ERD



This diagram shows the logical relationships among the entities, including basic relationship descriptions, cardinalities, and attributes.

3.2 Relational Schema

User (userID, fname, email, role, accountStatus, createdAt)

Listing (listingID, userID, title, description, price, condition, createdAt)

Category (categoryID, name)

Listing_Category (listingID, categoryID)

Review (reviewID, reviewerID, targetUserID, listingID, rating, comment, createdAt)

Message (messageID, senderID, receiverID, listingID, content, messageStatus, sentAt)

Report (reportID, reason, reporterID, targetUserID, listingID, createdAt)

Transaction (transactionID, price, buyerID, sellerID, listingID)

Favorites (userID, listingID, createdAt)

3.3 Normalized Relations

First Normal Form

All relations satisfy First Normal because each attribute contains a single value; each relation has a primary key, and there are no repeating groups or multivalued attributes.

Second Normal Form

All relations satisfy Second Normal Form because they are in First Normal Form, and all non-key attributes depend on the complete primary key. Relations with composite primary keys include:

- Listing_Reaction (userID, listingID)
- Listing_Category (listingID, categoryID)
- Review_Reaction (reviewID, userID)
- Favorites (userID, listingID)

Third Normal Form

Message (messageID, conversationID, senderID, receiverID, listingID, content, messageStatus, sentAt)

By getting rid of transitive dependencies in Conversation relation and Message relation, all relations satisfy Third Normal Form because they are in Second Normal Form and contain no transitive dependencies. Non-key attributes depend only on the primary key of their relation.

3.4 Primary Keys and Foreign Keys

Table 1: Primary and Foreign Keys

Relation	Primary Key	Foreign Key
User	userID	-
Listing	listingID	userID
Listing_Reaction	userID, listingID	userID, listingID
Category	categoryID	-
Listing_Category	listingID, categoryID	listingID, categoryID
Listing_Image	imageID	listingID
Review	reviewID	reviewerID(userID) targetUserID(userID) listingID
Review_Reaction	reviewID, userID	reviewID, userID
Conversation	conversationID	listingID buyerID(userID)
Message	messageID	conversationID, senderID(userID)
Report	reportID	reporterID(userID), targetUserID(userID), listingID
Favorites	userID, listingID	userID, listingID

3.5 Relationship Mapping

The relationships identified in the conceptual data model were mapped into the relational model using primary and foreign keys.

One-to-Many Relationships

- One user can create many listings, while each listing belongs to one user
- One listing can have many images, while each image belongs to one listing
- One listing can have many reviews, while each review is associated with one listing
- One user can write many reviews
- One user can receive many reviews
- One user can send many messages
- One review can have many replies
- One user can submit many reports

These relationships are implemented by placing the primary key of the parent relation as a foreign key in the child relation.

Many-to-Many Relationships

The many-to-many relationships were converted into associative relations:

- Users and listings have a many-to-many favourite relationship through Favorites.
- Users and listings have a many-to-many reaction relationship through Listing_Reaction.
- Listings and categories have a many-to-many relationship through Listing_Category.
- Users and reviews have a many-to-many reaction relationship through Review_Reaction.

4.2 Database Tables

The names of the Database tables are as follows:

- ***User***
Stores the information about the users including userID (pk), first name, role, email, account status, account creation date and time.
- ***Messages***
Includes the information regarding the messages exchanged, including the messageID (pk), content of the message, recieverID, listingID and senderID.
- ***Favorites***
The Favorites table includes information about the listings favored by the user. It includes the listingID, userID and createdAt.
- ***Listing***
The Listing table includes the information regarding the listings. It includes the listingID (pk), title and description of the listing, price, userID of the user who created the listing, and the date and time of creation.
- ***Review***
The Review Table includes information about the reviews submitted by a user. It includes the reviewID (pk), rating, comment, listingID, targetUserID, reviewerID and the date and time of the review.
- ***Report***
The Report table includes the information about the reports submitted by the user. It includes the reportID(pk), reason for the report, date and time of the report, reporterID, targetUserID and listingID.
- ***Transaction***
The Transaction table includes information about the transactions between the users. It includes the transactionID (pk), price, buyerID, sellerID and the listingID.
- ***ListingCategory***
The ListingCategory table includes the information about the category a listing belongs to. It includes the categoryID, listingID, and createdAt.
- ***Category***
The Category table includes information about the categories a listing can belong to. It includes the categoryID (pk) and the name of the category.

4.3 Field Definitions and Data Types

Table 2: User Attributes

Field	Definition	Data Type
UserID (pk)	The ID of the user	INT
Fname	First name of the user	VARCHAR (50)
Role	Authorization of the user	VARCHAR (20)
Email	KPU email of the user	VARCHAR (100)
accountStatus	Current status of user	VARCHAR (20)
createdAt	Creation date and time of the user account	DATETIME

Table 3: Message Attributes

Field	Definition	Data Type
messageID	The ID of the message	INT
Content	The contents of the message	TEXT
createdAt	Date and time of the message sent	DATETIME
receiverID	The userID of the message receiver	INT
listingID	The ID of the listing regarding which the messages were exchanged	INT
senderID	The userID of the message sender	INT

Table 4: Favorites Attributes

Field	Definition	Data Type
listingID	The ID of the listing added to the favorites by the user	INT
userID	The userID	INT
createdAt	The date and time of the listing added	DATETIME

Table 5: Listing Attributes

Field	Definition	Data Type
listingID	The ID of the listing	INT
Title	The title of the listing	VARCHAR (255)
Description	The description of the listing	TEXT
Price	The price of the item listed	DECIMAL (10.2)
createdAt	The date and time of the listing created	DATETIME
userID	The userID of the user who created the listing	INT

Table 6: Review Attributes

Field	Description	Data Types
reviewID	The ID of the review	INT
Rating	The rating given by the buyer to the listing	INT
Comment	The comment written by the reviewer	TEXT
createdAt	The date and time of the review written	DATETIME
listingID	The ID of the listing	INT
targetUserID	The userID of the user about whom the review is	INT
reviewerID	The userID of the reviewer	INT

Table 7: Report Attributes

Field	Description	Data Types
reportID	The ID of the report	INT
Reason	The reason for the report mentioned by the reporter	TEXT
createdAt	The date and time at which the report was submitted	DATETIME
reporterID	The userID of the reporter	INT
targetUserID	The userID of the user against whom the report is	INT
listingID	The listingID of the listing for which the report is	INT

Table 8: Transaction Attributes

Field	Description	Data Type
transactionID	The unique ID of the transaction	INT
Price	The price of the listing	DECIMAL (10.2)
buyerID	The userID of the buyer	INT
SellerID	The userID of the seller	INT
listingID	The ID of the listing	INT

Table 9: ListingCategory Attributes

Field	Description	Data Types
categoryID	The category to which the listing belongs	INT
listingID	The ID of the listing	INT
createdAt	The date and time of the listing added	DATETIME

Table 10: Category Attributes

Field	Description	Data Type
categoryID	The unique ID of the category	INT
Name	The name of the category	VARCHAR (100)

4.4 Constraints and Integrity Controls

The constraints and integrity controls that keep the database valid, consistent and accurate are:

- Primary Key constraints
 - Ensure each record in the table has a unique identifier. The primary key cannot be null.
- Foreign Key Constraints
 - Maintains relationships between tables.
- NOT NULL constraints
 - Prevents the mandatory fields from being left empty.
- CHECK constraints
 - Restricts the values to valid range or conditions.

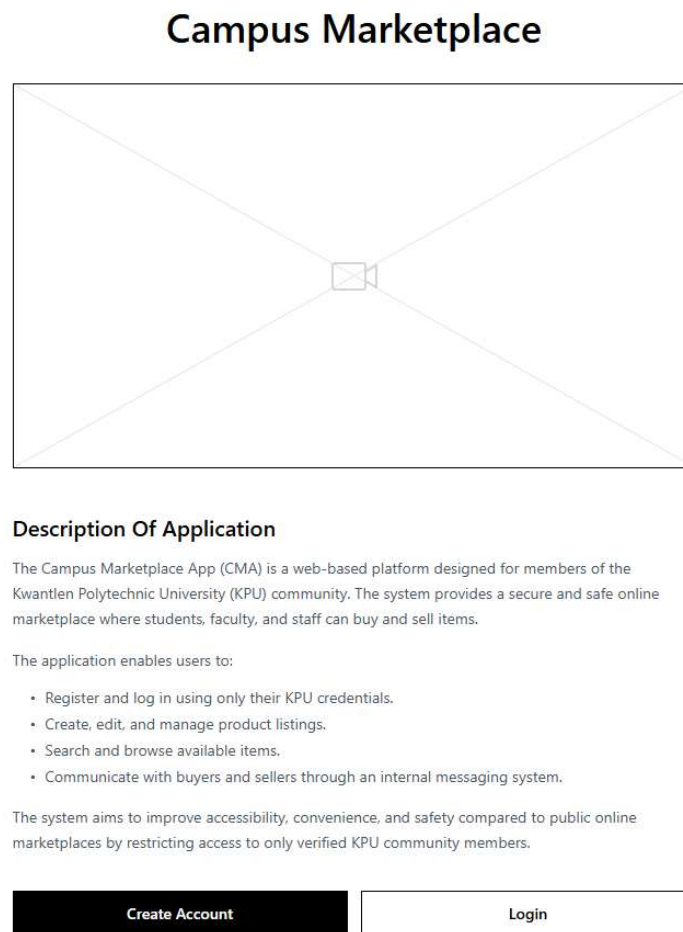
5. User Interface Design

This section presents the UI design of the Campus Marketplace Application. The wireframes are created with Figma.

5.1 UI Design Overview

The UI design is centered around eight key functions. The following sections illustrate and describe the UI for these functions.

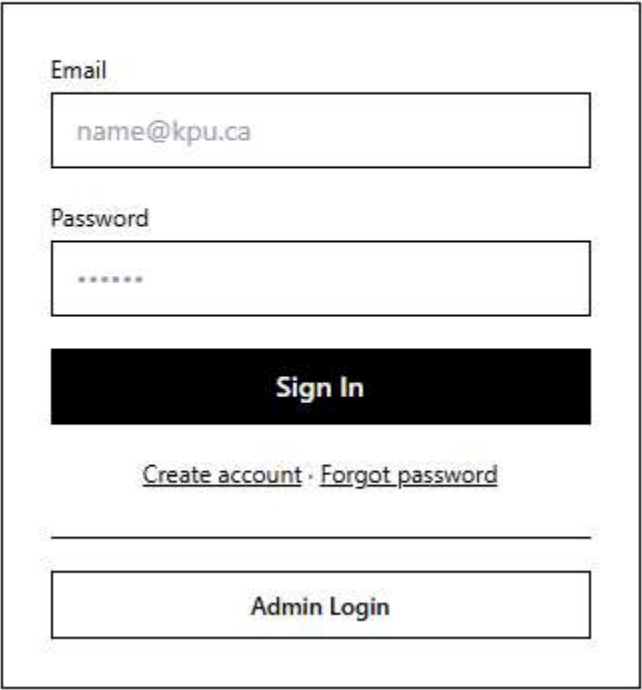
Figure 5.1: The Welcome Page



5.2 Login Form

The Welcome page of the Campus Marketplace Application displays a brief description of the Application. The Login page can be accessed by clicking the Login button at the bottom of the Welcome page. The user will be promoted to the Login page.

Figure 5.2: Login page



The login form is titled "Login" with the subtitle "kpu.ca accounts only". It contains the following elements:

- An "Email" label above a text input field containing "name@kpu.ca".
- A "Password" label above a text input field containing six asterisks "*****".
- A black "Sign In" button.
- Links for "[Create account](#)" and "[Forgot password](#)".
- A horizontal line separating the main login section from the "Admin Login" button.

The login page displays the following options:

- Text boxes to enter the KPU email and the password to the accounts to login.
- A login button that needs to be pressed once the credentials are filled.
- Create Account option for new users that wish to create an account.
- Forgot Password option that enables the users to reset their password using their KPU email.
- Admin Login option that lets the administrators log in into their administrator accounts with their credentials.

5.3 Registration Form

Figure 5.3: Registration Page

Create Account
@kpu.ca email required

Full Name

KPU Email

Password

☐ I agree to terms of service

Sign Up

Already registered? [Login](#)

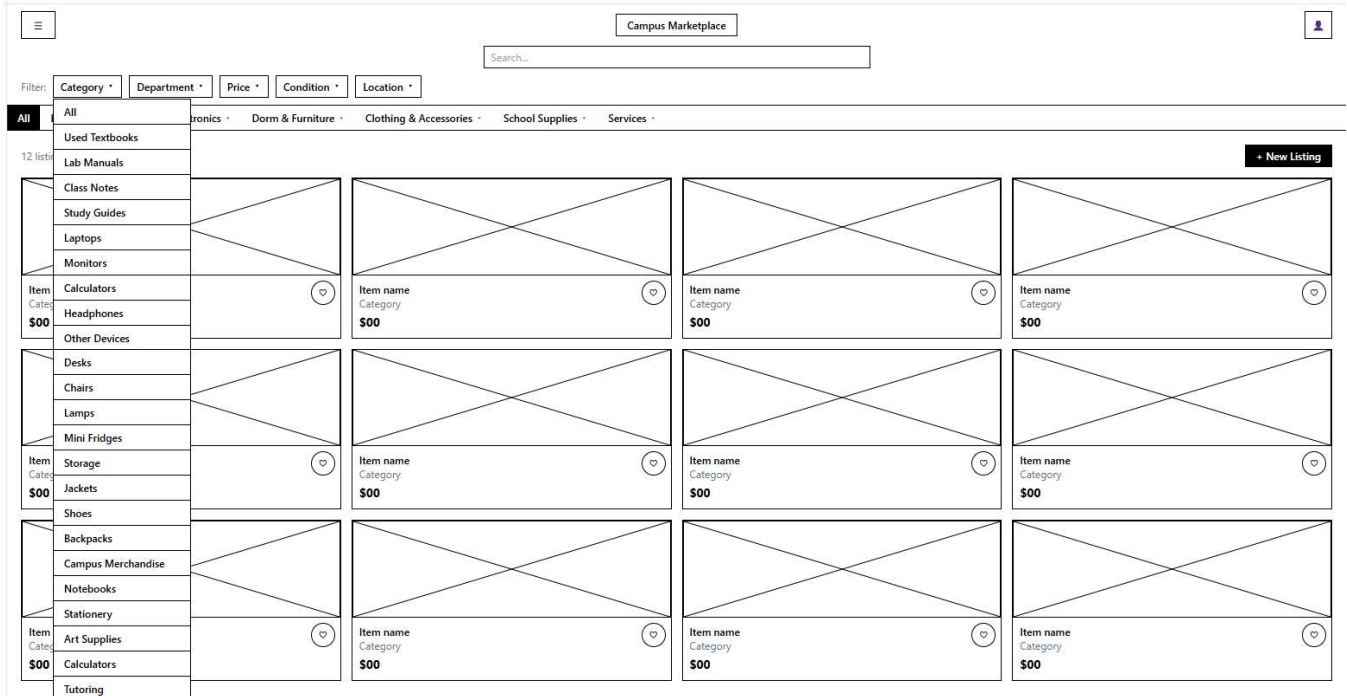
The registration page displays the following options:

- Text boxes to enter the KPU email, password, and preferred name.
- A check box to confirm that the user agrees with the protocol.
- A sign-up button that needs to be pressed once the credentials are filled.
- Login option if the user has already registered.

5.4 Browse Listings Form

The menu option of Browse Listings enables the user to browse the listings. The search bar at the top of the Browse Listings Form enables the user to search for the listings they are interested in.

Figure 5.4: Browse Page



The user can apply the following filters while looking for the listings:

- Category
 - The category refers to the type of listings, e.g. textbooks, clothes, electronics, etc.
- Department
 - The department of the listing refers to the academic department the listing belongs to. E.g. business, nursing, criminology, etc.
- Price
 - The filter Price enables the user to view the listings in the price range they desire. E.g. Under \$10, \$10-\$50, \$50-\$100, etc.
- Condition
 - This filter helps the user to filter results according to the condition of the product. E.g. New, Like new, Good and Fair.
- Location
 - The user can filter the location of the listing by the KPU campus locations. E.g. Surrey Campus, Langley Campus, etc.

Apart from this, some additional filters are provided:

- All
 - Let the user view all the listings without any filters applied.
- Books and Textbooks

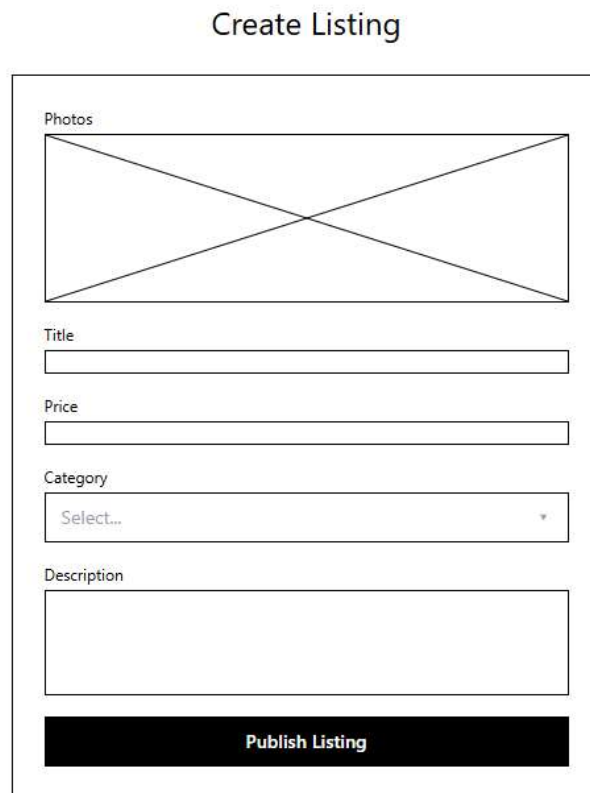
- The drop-down menu shows options like used textbooks, lab manuals, etc. The user can choose what types of books listing they want to view.
- Electronics
 - The drop-down menu shows the options for different electronic devices.
- Dorm and Furniture
 - The drop-down menu gives users options such as desks, chairs, etc.
- Clothing and Accessories
 - The drop-down menu displays options such as shoes, campus merchandise, jackets, etc.
- School Supplies
 - The drop-down menu displays options such as notebooks, stationery, art supplies, calculators, etc.
- Services
 - The drop-down menu displays the options such as tutoring, editing, rideshare and other Services.

5.5 Create Listing Form

The Create Listings form enables the user to create a listing of the items they desire to sell.

Figure 5.5: Create Listing Page

Create Listing



Photos

Title

Price

Category

Select...

Description

Publish Listing

The prompt form displays the following options that are required to be filled in to successfully create a listing:

- File uploading control to upload listing image.
- Text boxes to enter the title, price, and description of the listing.
- Dropdown list to select categories of the listing.
- Button that needs to be pressed once the credentials are filled.

5.6 Listing Detail Form

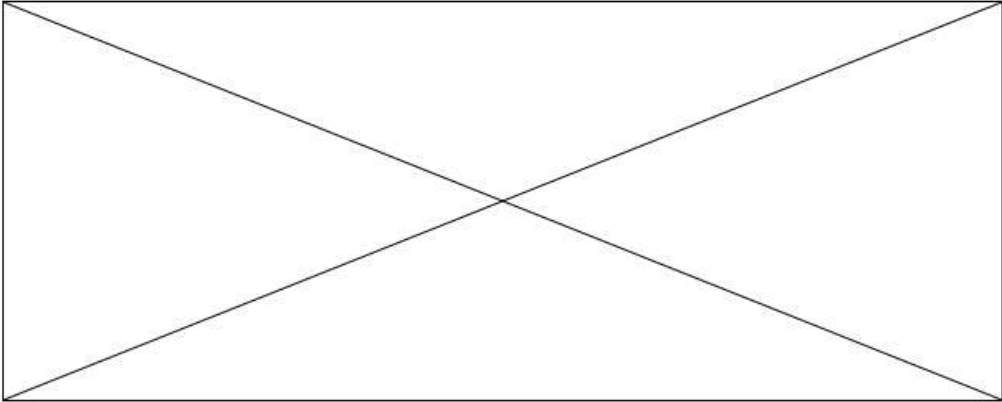
The listing detail page enables the user to check the detail description of an item and to take further action.

Figure 5.6: Listing Detail Page

Campus Marketplace

Search...

[← Back to listings](#)



Item name

Category · Posted 0 days ago

\$00



Item description goes here. Condition, details, and any other notes about the item would be listed in this space.

[Report listing](#)



Seller name

Department · Rating 0.0

[Report seller](#)

Contact Seller

The user can see following information about the item:

- Basic Info
 - Full size image, item name, price, description
- Additional Info
 - Categories list, release date, favorite list saving status.
- Seller Info
 - Seller name, seller department, Buyer review on the seller.
- Report button
 - Report the listing or the seller.
- Contact Seller
 - Button to start a chat with the seller.

5.7 Messaging Form

The messaging form enables the user to communicate with the seller. The chat record will be shown in the center area.

Figure 5.7: Chat Page

The diagram illustrates the layout of the chat page. It features a top navigation bar with the 'Campus Marketplace' logo. Below this is a search bar. A horizontal line divides the page into a header section and a main content area. The main content area includes a 'Back to listing' link. Below the link is a box for the chat header, which contains a close button (X), the seller's name, and the item name. The central part of the page is a large chat area with two 'Messaging...' buttons. At the bottom of the chat area is a text input field labeled 'Type a message...' and a 'Send' button.

The chat page displays the following options:

- Text box to enter the message
- Button to send the message to the seller/user
- The messages exchanges, if any
- The name of the seller along with the item name

5.8 User Profile

The user profile page enables users to check and modify self-related information.

Figure 5.8: User Profile Page

Campus Marketplace

Search...



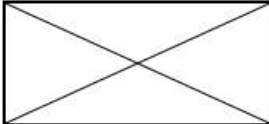
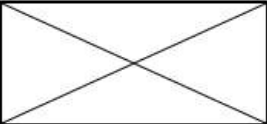
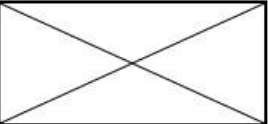
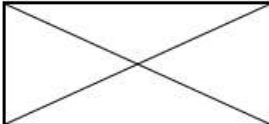
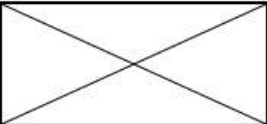
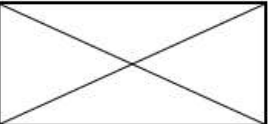
Name
name@kpu.ca
Department · Year
Listings: 00 Sold: 00 Rating: 0.0

Log out

Listings

Saved

Settings

 Item name \$00	 Item name \$00	 Item name \$00
 Item name \$00	 Item name \$00	 Item name \$00

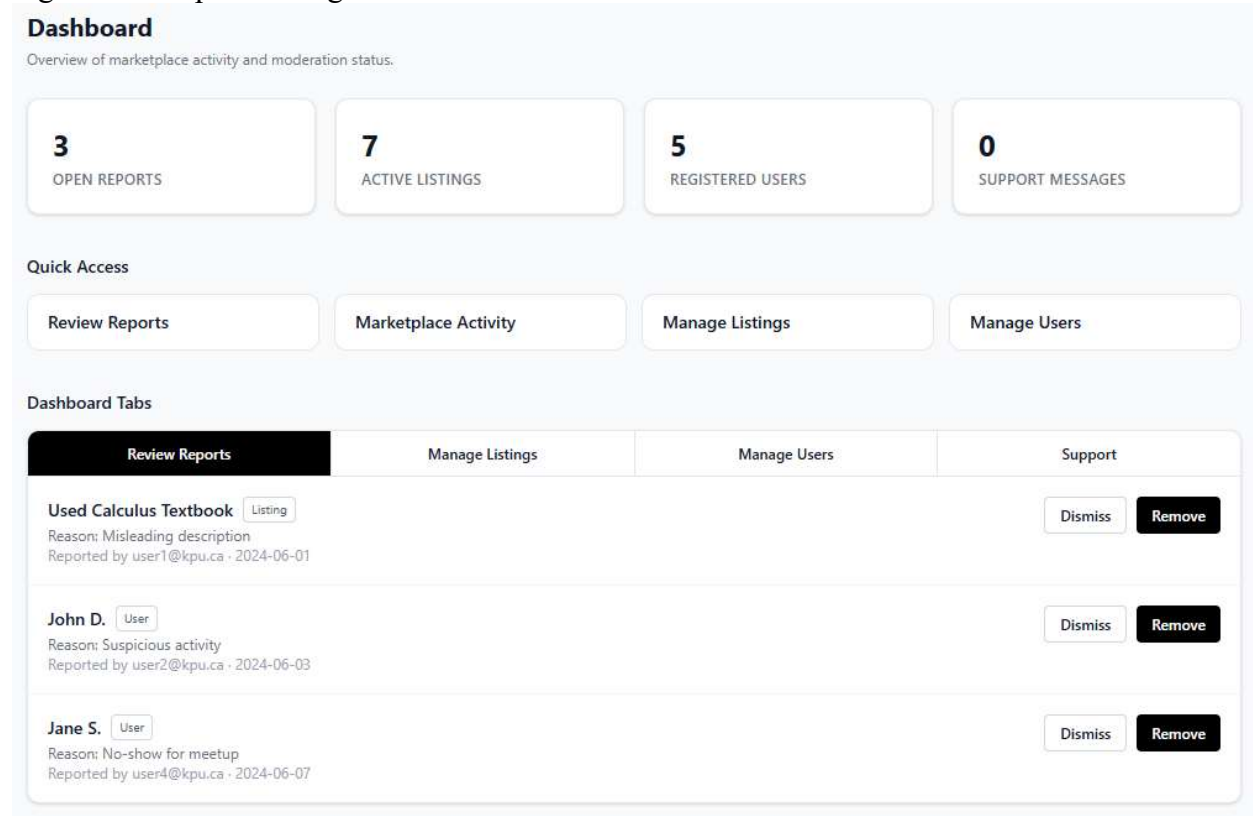
The user can see following information:

- Personal Info
 - Profile photo, name, department, e-mail address, grades, and log out options.
- Listing Info as a seller
 - Listing count, sold count, review rating from buyer, Listing detail drop-down list.
- Saved Info as a buyer
 - Saved listing details drop-down list
- Setting
 - Fields to change personal info, such as display name, password, etc.

5.9 Admin/Report Management Form

The management dashboard enables administrators to track reports, manage listings & users, and statistic the data.

Figure 5.9: Report Management Form



The administrators can do following actions:

- Dealing reports
 - Review the details of report, processing status of report, and executable operations
- Manage listings
 - Number of active listings, listing details, and executable operations
- Manage users
 - Number of registered users, user detail, and executable operations
- Support
 - Messages sent from user

6. Reports

Reports from the system share a perspective, such as statistics relating to listings and users. Administrators are the ones who can see the statistical data, make decisions to benefit the platform and keep it secure and engaging for the user.

6.1 Listing Activity Report

The listing activity reports tell the administrators details about the most favored categories, price range, etc. Depending upon the statistics and reports the administrators can take appropriate actions. The administrators can choose to subdivide the category to make the experience more efficient for the user, or the administrators can choose to invest in the algorithm to push the most favored listings or categories to more users.

6.2 User Activity Report

The user activity report tells the administrators about information such as the most active users and the users with the highest number of active listings.

7. Conclusion

This project design outlines a structured and well-organized database structure that meets the requirements of the application. The conceptual, logical, and physical diagrams depict clarity in data organization, ensuring proper relationships between entities and data integrity through appropriate constraints. The design is scalable, efficient, and well-organized, which enables it to be easily modifiable in the future according to the needs. Since the design meets the requirements of the application, it is ready to be implemented.